Verification of Nyquist–Shannon Sampling Theorem

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Deceleration

I, the undersigned, **Sreedoy Sarker**, hereby declare that the work presented in this report is my own, carried out under the scientific direction of **Dr. Md Rasedujjaman**, in accordance with the principles of honesty, integrity, and responsibility. This report was written entirely by myself and complies with the charter on the fight against scientific plagiarism.

Sylhet, January 18, 2026

Signature

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Experiment Number: 01

Name of the experiment: *Verification of Nyquist–Shannon Sampling Theorem*

1 Objective

The primary objective is to validate the Nyquist-Shannon Sampling Theorem by analyzing its application to a multi-frequency composite signal. Specifically, the experiment aims to:

Determine the effective Nyquist rate for a signal containing multiple frequency components.

Demonstrate "partial aliasing," where high-frequency components are distorted while low-frequency components remain intact.

Verify signal reconstruction accuracy using Python-based simulation.

2 Introduction

In digital communication and signal processing, real-world signals (like voice or music) are rarely single pure tones; they are complex sums of various frequencies. The Nyquist-Shannon Sampling Theorem dictates that to digitize these signals without loss, the sampling frequency (f_s) must be at least twice the maximum frequency present in the spectrum (f_{max})11. Failure to meet this criterion leads to aliasing, where high-frequency details are folded into lower frequencies, creating irreversible distortion22. This experiment utilizes a composite signal to visualize these effects dynamically

3 Theoretical Analysis

3.1 The Sampling Theorem for Composite Signals

For a signal $x(t)$ composed of multiple frequencies $f_1, f_2, \ldots, f_n$, the signal is band-limited by the highest constituent frequency:

$$f_{max} = \max(f_1, f_2, \ldots, f_n)$$

To prevent aliasing of any component, the sampling frequency must satisfy:

$$f_s \geq 2f_{max}$$

If $2f_{lower} < f_s < 2f_{higher}$, the lower frequency component will be captured correctly, but the higher component will alias. This results in a reconstructed signal that bears no resemblance to the original waveform.

3.2 Reconstruction

Reconstruction attempts to recover $x(t)$ from samples $x[n]$. While ideal reconstruction uses Sinc interpolation 3, practical systems often use linear interpolation4. For composite signals, linear interpolation requires significant over-sampling ($f_s \gg 2f_{max}$) to accurately trace the complex interference patterns between the multiple frequencies.

4 Experimental Setup & Tools

Platform: Python 3.x (NumPy for calculus, Matplotlib for visualization).

Hardware: Standard Computing Workstation.

Signal Model: A composite sum of two sinusoidal waves with distinct frequencies.

5 Methodology

We define a continuous-time composite signal $x(t)$ comprising a low-frequency base and a high-frequency detail:

$$x(t) = \sin(2\pi f_1 t) + \sin(2\pi f_2 t) \tag{1}$$

The signal parameters are defined as follows:

- **Frequency 1 (f_1):** 5 Hz
- **Frequency 2 (f_2):** 12 Hz
- **Maximum Frequency (f_{max}):** 12 Hz
- **Theoretical Nyquist Rate:** $2 \times 12 = 24$ Hz.

We will test three sampling regimes:

- **Under-sampling (15 Hz):** Sufficient for f_1 ($15 > 10$), but insufficient for f_2 ($15 < 24$). This will cause partial aliasing.
- **Critical Sampling (24 Hz):** The mathematical limit.
- **Over-sampling (100 Hz):** High fidelity reconstruction.

6 Python Implementation

The following script generates a composite analog signal and simulates sampling at three distinct frequencies ($f_s = 15, 24, 100$ Hz) to demonstrate the effects of aliasing compared to the Nyquist rate.

Listing 1: Composite Signal Sampling Simulation

```python
import numpy as np
import matplotlib.pyplot as plt

# --- 1. Define Composite Signal ---
# We use a mix of 5Hz and 12Hz. f_max is 12Hz.
f1 = 5
f2 = 12
f_max = max(f1, f2)      # 12 Hz
nyquist_rate = 2 * f_max # 24 Hz

# Time setup
duration = 0.5 # seconds
t_continuous = np.linspace(0, duration, 1000)

# The "True" Analog Signal: Sum of two sine waves
x_continuous = np.sin(2 * np.pi * f1 * t_continuous) + \
               np.sin(2 * np.pi * f2 * t_continuous)

# --- 2. Define Sampling Scenarios ---
# Scenario A: 15 Hz (Under-sampling: Fails for 12Hz component)
# Scenario B: 24 Hz (Critical Sampling)
# Scenario C: 100 Hz (Over-sampling)
fs_rates = [15, 24, 100]
labels = ["Under-sampling (15 Hz)",
          "Critical Sampling (24 Hz)",
          "Over-sampling (100 Hz)"]
colors = ['red', 'orange', 'green']

plt.figure(figsize=(10, 12))

```

```python
for i, fs in enumerate(fs_rates):
    # Generate Samples
    t_sampled = np.arange(0, duration, 1/fs)

    # Sample the composite function
    x_sampled = np.sin(2 * np.pi * f1 * t_sampled) + \
                np.sin(2 * np.pi * f2 * t_sampled)

    # Plotting
    plt.subplot(3, 1, i+1)

    # Ground Truth (Gray)
    plt.plot(t_continuous, x_continuous, 'gray', alpha=0.5,
             linewidth=2, label='True Composite Signal')

    # Samples (Stem)
    plt.stem(t_sampled, x_sampled, linefmt='b',
             markerfmt='bo', basefmt=" ", label='Samples')

    # Reconstruction (Linear Interpolation)
    plt.plot(t_sampled, x_sampled, color=colors[i],
             linestyle='--', linewidth=2, label='Reconstructed')

    # Status Logic
    if fs < nyquist_rate:
        status = "FAILED: High Freq Aliased"
    elif fs == nyquist_rate:
        status = "MARGINAL: Phase Sensitive"
    else:
        status = "SUCCESS: Accurate"

    plt.title(f"{labels[i]} | {status}")
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.legend(loc='upper right', fontsize=8)

plt.tight_layout()
plt.show()
```

7 Results and Analysis

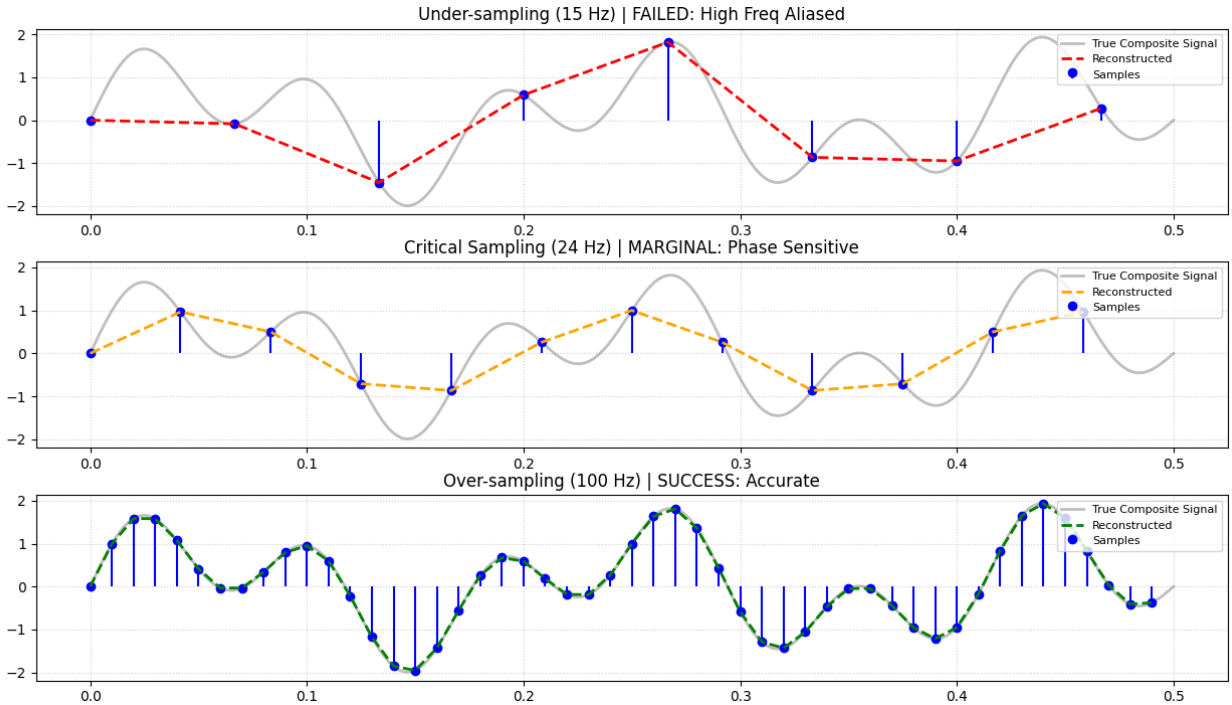

Figure 1: Time-domain reconstruction of a composite signal ($f_{max} = 12$ Hz) under three sampling regimes: under-sampling (15 Hz), critical sampling (24 Hz), and over-sampling (100 Hz).

Observation at 15 Hz ($f_s < 2f_{max}$):The reconstruction (red dashed line) failed significantly. While the 5 Hz component was theoretically captured, the 12 Hz component was undersampled. The result is a distorted wave that does not match the peak locations or the rhythm of the original gray signal. The complex "ripples" of the 12 Hz wave were lost.

Observation at 24 Hz ($f_s = 2f_{max}$):At the critical rate, the sampling points hit specific repeating parts of the wave. While the frequency content is technically preserved, the visual shape is a jagged triangle. The amplitude at the peaks is inconsistent because the samples do not always align with the signal maxima.

Observation at 100 Hz ($f_s \gg 2f_{max}$):The green dashed line traces the complex "double-hump" shape of the composite signal effectively. Both the slow 5 Hz oscillation and the fast 12 Hz ripple are clearly visible and accurately reconstructed.

8 Discussion

This experiment verified that for a composite signal, the sampling rate is dictated strictly by the highest frequency component. In our under-sampling case ($f_s = 15$ Hz), we observed a phenomenon known as *partial aliasing*.

- The 5 Hz component required $f_s \geq 10$ Hz. Our 15 Hz rate satisfied this condition.
- The 12 Hz component required $f_s \geq 24$ Hz. Our 15 Hz rate violated this condition.

Consequently, the 12 Hz component "folded" and corrupted the clean 5 Hz signal, resulting in a waveform that contained false information. This proves that one cannot simply sample at the average frequency; the Nyquist limit is an absolute threshold based on the signal's maximum bandwidth.

Furthermore, linear interpolation proved insufficient at the critical rate (24 Hz), as the complex curvature of a composite signal requires a much higher sample density to be visually resolved.

9 Conclusion

The experiment successfully verified the Nyquist-Shannon Sampling Theorem using a multi-frequency signal. We concluded that the sampling frequency must be at least twice the maximum frequency component (f_{max}) of the composite signal. Failure to meet this for even one component results in aliasing that distorts the entire signal reconstruction. For practical applications using linear interpolation, over-sampling is essential to preserve waveform fidelity.

10 References

[1]A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ: Pearson, 2010, pp. 115–135.

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